Learning, Forgetting and Recovery in Evolved Agents with a Frozen Language-Model Prior: Preregistered Studies, Including the Negative Results

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DRAFT — November 1, 2026

Abstract

We study simulated foragers whose decision layer is evolved (innate), whose beliefs are learned within a lifetime (acquired), and whose perception of foods described only in natural language comes from a frozen pretrained decision model (Laya; perceptual). Across twenty-one preregistered evaluations we report four groups of findings. *(1) The frozen prior pays only in some worlds*: its fitness value grows with consequence delay (+0.049 at delay 40, +0.073 at delay 80, 8/8 seeds), is about zero under immediate feedback, and is zero in a world simple enough for experience-based learning. *(2) Retention.* After the rules reverse and later return, an ordinary learner is worse the second time it meets the old world (savings −0.12), whereas an environment-table memory that opens a new belief table on prediction error saves +0.13; this replicated on 16 fresh training seeds, on an independent data source (UCI Mushroom; there Laya's prior has AUC 0.505, so the gain does not depend on it) and with twice as many foods (+0.25, +0.21, +0.14 and +0.37 over the ordinary learner in four conditions, 16/16 seeds). *(3) Language as evidence.* Asking an LLM to decide when to switch memory did not help; but caching an LLM's judgement of whether two cue sentences describe the same conditions and fusing it as a noisy sensor into Bayesian context inference gave savings +0.25 against −0.11 with a never-repeating cue wording (16/16 seeds), nearly equal to an oracle sensor, and collapsed when the sensor was shuffled; with 30% of judgements inverted its advantage disappeared. *(4) Reliability.* Evolving in a history that includes unreliable cues made the innate trust in a cue drop from 0.99 to 0.91 and raised fitness by +0.24 when cues misled (and from 0.26 to 0.82 with no cue at all), at a cost of 0.009 when cues were reliable; learning the reliability within a lifetime did *not* add value (−0.03), and a second independent channel helped less than preregistered (+0.047 against +0.05). The individual-level advantage did not carry over to reproducing populations, and a heritable-mutation probe found no selection direction. We report all negative results and the limits of the design.

1 Introduction

Pretrained language models carry broad world knowledge, and recent work uses them as priors or guides for decision-making agents [17, 5, 2]. Whether such a prior is actually worth having is rarely isolated: a gain may come from the model's knowledge, from the agent's own learning, from a better-tuned decision rule, or from any informative signal at all, including a simple lexicon.

We study this question in a setting where the three possible sources can be pulled apart. Agents are produced by evolution, which sets an innate decision layer; they learn within a lifetime, which

sets acquired beliefs; and they perceive the world partly through a frozen pretrained model that acts as a fast "System 1" intuition. The frozen model is never trained and its weights are not evolved, so the innate genome, the acquired beliefs and the perceptual prior are separately attributable. The world supplies the problem worth learning: foods are described only in natural language, and the simulator can add delayed consequences, lethal items, deceptive descriptions, social cues and ripeness cycles, each off by default.

Because learning can itself make a prior redundant, and because an early large effect in our own work did not survive repetition (Section 5), we preregistered each confirmatory evaluation, evaluated on held-out dialects, repeated training seeds, and report every result including the unsupported hypotheses. The paper characterises *when* the prior pays rather than asserting that it does.

A second question is closer to what a lifetime learner is for: can an individual learn from consequences, adapt when the environment changes, and keep or recover what it knew when the old environment returns? Ordinary incremental learners overwrite old knowledge when the world changes. We test whether an innate memory structure evolved together with the decision layer, and requiring no change to the frozen model, fixes this, whether it helps when agents reproduce, and whether the frozen model can supply the missing context signal.

Contributions (claims supported by preregistered evidence only).

1. A ladder of world complexity with gates (generated ontology; lethal, delayed and deceptive consequences) in which innate, acquired and perceptual contributions can be separated.
2. Evidence that a fixed model discriminates unseen phrasings and knowledge-dependent descriptions without fine-tuning, together with its option-order bias and our handling of it.
3. A characterisation of *when* the prior pays: delay, deception, and learnability.
4. A measure of retention (savings) in reversal worlds and evidence that an environment-table memory evolved by evolution strategies, switching on prediction error without any cue, turns negative savings of an ordinary learner into positive savings, including under partial, repeated and long reversals not seen in training.
5. Negative and bounding results: the individual-level advantage did not produce a detectable population-level advantage in reproducing populations; a heritable-mutation probe found no detectable selection direction; cue-keyed memory depends on the cue being present; the frozen model did not read environment cues in two preregistered-style probes.
6. A Bayesian context-inference learner in which a frozen LLM's cached judgement of cue sentences enters as a noisy sensor (not a command); evidence that this beats an environment-table memory when cue wording never repeats, that the gain comes from reading language, and that it breaks when the sensor errs.
7. A test of the division of labour for cue reliability: the innate prior must be evolved in a history that includes unreliable cues; lifetime re-estimation did not help in our design.
8. Methodology: preregistration, held-out and independently sourced data, repeated seeds, and honest reporting of non-replication.

Not claimed: that the prior reliably raises fitness in general; that Laya beats a lexicon on fitness in general; that Laya understands environmental change; that an LLM is a reliable decision-maker here; that lifetime reliability estimation helps; that memory structure is favoured by selection; any "first" claim.

2 Related Work

Language-model priors for decision making. Yan et al. [17] treat a language model as a prior over actions and combine it with reinforcement learning by Bayesian inference, reporting large gains in sample efficiency. Du et al. [5] use a language model to suggest goals that shape exploration during pretraining, and Ahn et al. [2] ground a language model's proposals in the value functions of pretrained robotic skills so that suggestions are feasible. These works ask how to use the model's knowledge; we ask when a frozen model's prior is worth anything once the decision layer is evolved and the agent also learns, and we include a lexicon baseline and a shuffled-prior control.

Evolution and lifetime learning. Hinton and Nowlan [10] showed in simulation that learning can guide evolution, a result usually described as the Baldwin effect; Fontanari and Santos [7] argue that this simulation demonstrated that the ability to learn is easily selected for, and give a direct demonstration of the effect. Later work evolves learners in richer settings: Gupta et al. [9] evolve embodied agent morphologies that learn control and report evidence of a morphological Baldwin effect; Le [13] combine evolution with self-teaching networks in a foraging simulation; Fernando et al. [6] evolve the initial parameters and hyperparameters of learning algorithms, so that selection acts on the starting point of learning, as in our genome; the abstract does not say whether weights learned within a lifetime are inherited, whereas in our setting acquired beliefs never are. In these works the starting point of learning is evolved or random; in ours a pretrained model supplies part of the agent's perception and stays fixed. We optimise the genome with evolution strategies [16]. Language models have also been used inside the evolutionary loop as mutation operators [14], a different role from the one studied here.

Forgetting and context. Neural networks trained on a sequence of tasks tend to overwrite earlier knowledge; Kirkpatrick et al. [12] reduce this by slowing learning on weights that mattered for earlier tasks. In animal learning, context-dependent behaviour such as the return of a response after extinction in a different context has been modelled by inferring latent causes [8]. Our environment-table learner is a simple heuristic in this spirit (a new belief table is opened when prediction error is high), with its few parameters evolved; we do not compare it with those methods, and we make no claim that it is optimal.

Lexicons and methodology. We use the VADER sentiment lexicon [11] as an independent prior that reads affect words but has no world knowledge. Our design separates hypothesis generation from testing by preregistration [15].

Scope of the search. Related work was collected by web search on 2026-10-21 (language-model priors for RL, Baldwin effect and evolved learners, evolution plus language models) and by reading each cited abstract; it is not exhaustive. We did not locate prior work that evolves agents on top of a frozen pretrained decision model while attributing innate, acquired and perceptual contributions separately, but we make no priority claim.

3 World and Method

World. The simulator owns all state; the model only proposes actions. Foods carry hidden attributes (nutritious, mildly toxic, strongly toxic, lethal) rendered into text by a generator. Lethal-

ity, delayed consequences and deception (description opposite to truth) are switchable and off by default. Hidden truth is used only to measure upper bounds and never enters an agent's input.

Three sources. (i) *Perceptual*: fixed Laya [3] returns an eat probability per description, cached by text; never trained. (ii) *Innate*: a genome of nine decision-layer genes (trust, curiosity, temperature, forgetting, retry, calibration slope/bias, risk threshold, delayed-credit window), evolved by evolution strategies; Laya weights are not evolved. (iii) *Acquired*: per-kind belief updates within a lifetime, with a credit window for delayed outcomes; not inherited.

Learners with context memory. Beyond the ordinary learner (*plain*), we evolve (a) *dual*: a fast and a slow belief set mixed by prediction error (13 genes); (b) *regime*: up to four environment belief tables, with a new table opened or an old one selected from prediction error, using three extra genes (opening threshold, rate, switching margin; 12 genes), no cue; (c) *cue*: beliefs keyed on a visible environment cue sentence used only as a label (9 genes). Beliefs start empty at birth and are never inherited. Individuals see only their own state, limited memory and local observations; the hidden environment state is used only by the observer.

Reversal worlds and savings. Six foods ($K = 6$; half nutritious, half toxic), hazard cap 1, 8 individuals, 1200 ticks per phase. State A is the original; B reverses every food's effect; C reverses a random half. Worlds are A/B/A, A/B/C, and, for boundary tests, half-reversal (A/B′/A), repeated (A/B/A/B/A) and long stay (A/B/B/B/A). The metric in a 200-tick window is the share of nutritious foods among foods eaten; *early* means the first two windows of a phase; *savings* = early share in the last A phase − early share in the first A phase. Fitness is as above. Agents were trained on A/B/A and A/B/C worlds with a true cue, with 64 independent evolutions (preregistration 17); boundary and population tests reuse the frozen genomes without retraining.

Reproducing populations. For the population test, the simulator's reproduction is switched on (supply 20, birth threshold 26, 16 initial individuals, maximum age 2400, cooldown 150; chosen by a degraded calibration rule because no candidate met the pre-written criterion). Reproduction is a fixed instinct, not a gene. A child inherits the parent's *innate type*; acquired beliefs start empty. Types are assigned by the test harness via lineage and are never an individual's input.

Independent data source. To avoid dialects written by the authors we also use the UCI Mushroom data [1] (CC BY 4.0): each food description is built from seven randomly chosen attributes of a real record (fixed random stream, 4000 descriptions), and the edible / poisonous label comes from the data. The frozen prior has no discriminating power on this pool (AUC 0.505, recorded before any individual was evaluated on it), so results there cannot come from the prior.

Bayesian context inference with a language sensor. The learner (*benv*, 14 genes: the nine base genes plus an observation noise, a context-change probability, a new-context share, a boundary change probability and a sensor reliability) keeps up to four belief tables; the posterior over "which context am I in" is maintained by forward filtering with conjugate normal predictive likelihoods whose variance shrinks with evidence, new contexts are opened from the prior predictive, tables are updated by posterior responsibility, and decisions use the posterior-weighted mixture. When the visible cue sentence changes, a sensor supplies a likelihood factor $f = r\,s + (1-r)(1-s)$, where s is the probability that the new sentence describes the same conditions as an earlier one of that context and r the sensor's reliability. s is an LLM's judgement (`deepseek-flash`, thinking off, served by

DeepSeek-V4.1-Flash-1017 [4]) of all 630 pairs of the 36 cue sentences, asked once in both orders and cached; training and evaluation look the cache up (synchronous, free, strictly replayable), and the individual sees no state labels. In the *paraphrase* world the same state is worded differently each time it returns.

Cue reliability regimes. In the dense-alternation world (all food effects reverse every 300 ticks, 4500 ticks, 14 boundaries) two cue channels (author-written and LLM-written sentences) are each independently in one regime unknown to the individual: reliable; noisy (30% of phases show the other state's sentence); uninformative (50%, independent of the state); or absent. The learner for this study (*benv3*, 15 genes) gives each channel a reliability with a Beta prior (innate mean and strength), optionally updated within a lifetime by checking, on the first three new foods eaten after a boundary, whether the environment the cue pointed to predicts them best; the factors of the channels multiply.

Conditions and metric. Prior $\in$ {Laya, none, shuffled Laya, VADER lexicon}. Fitness = $0.7\times$ survival fraction $+0.3\times$ final cohort energy fraction. The evaluated cohort is eight individuals without reproduction (except in the population test); the number of food kinds K and the hazard cap differ by protocol and are given in each preregistration (for example $K = 32$, cap 4 in preregistration 07; $K = 6$, cap 1 in preregistrations 17–19). Each evaluation uses new world seeds not used in training.

4 Evaluation Design

Gates G1–G6 for world complexity; preregistered protocols (`docs/PREREGISTRATION_02–22`) frozen by git commit before evaluation; held-out dialects written blind to training dialects; independent training seeds; intersection tests with Holm correction where stated. Directional confirmation and magnitude confirmation are reported separately (Table 1).

5 Results

5.1 An early large effect did not replicate

In preregistration 02 (one training seed), evolved Laya beat no-prior by +0.144 [0.119, 0.168] and shuffled prior by +0.155, but not VADER (+0.012 [−0.009, 0.033]; H3 not supported). With three training seeds (preregistration 03) H1–H3 all failed the three-seed criterion (H1 per-seed: +0.150, +0.011, +0.039). The large effect came mainly from evolutionary bias in the no-prior and shuffled arms of seed 1.

5.2 Value depends on delay

5.3 Prior versus lexicon

On a fact dialect (neutral tone: related species, origin, animal behaviour), Laya AUC is 0.775 versus 0.583 for VADER ($\Delta = +0.192$ [0.170, 0.212]). In fitness, Laya−VADER is +0.050 (8/8 seeds, delay 40). Social observation of neighbours' visible appearance gives +0.025 (8/8; wording B, unseen in training). But on ripeness-stage phrases a lexicon can read, Laya−VADER is +0.005 (not supported). The difference converts to fitness only when understanding is required.

Prereg	Comparison	Mean	95% CI	Verdict
06	delay 0, Laya−none (non-inferiority > −0.04)	+0.0154	[−0.0076, +0.0520]	supported
06	delay 40, Laya−none (matched training)	+0.0486	[+0.0248, +0.0709]	supported
07	held-out, delay 80, Laya−none	+0.0732	[+0.0691, +0.0765]	supported
07	value grows with delay (d80−d40)	+0.0350	[+0.0247, +0.0459]	supported
06	hierarchical belief, deception 0.40	+0.0005	[−0.0033, +0.0043]	not supported

Table 1: Fitness differences (seed-level bootstrap CI), 6–8 training seeds. Source: `docs/EFFECT_SIZES.md`. Under mismatched train/eval regimes (prereg 04) the prior appeared harmful at delay 0; matched training removed this (+0.015), so that harm was a train–eval mismatch artefact.

5.4 When the world is learnable the prior is worth nothing

With $K = 4$ kinds (preregistration 11) Laya−none = −0.0049 (threshold +0.03; not supported). The no-prior agents rarely refuse to eat (12.5% vs. 75% at $K = 16$), fitness 0.315 against a blind rule 0.162 and an oracle 0.429: the first fair comparison, and the prior's net value is zero. The no-prior agents gain from finer observation granularity (ripeness phrases), Laya agents from semantics; the two mechanisms cancel.

5.5 Transfer

Trained on the fact dialect, Laya agents transfer to the held-out dialect (Laya−VADER +0.0621, 8/8) while lexicon agents do not.

5.6 Retention: an environment-table memory saves, an ordinary learner does not

The ordinary learner is worse when A returns than the first time it met A, because experience in B has overwritten its beliefs. A learner that infers the environment from prediction error and keeps separate tables turns this negative savings into a positive one, and the genome evolved a low threshold, i.e. it opens new tables readily. The advantage holds in worlds that differ from training: partial reversal, repeated reversal, and a long stay in B (savings +0.07, +0.15, +0.18 against about zero or negative for the ordinary learner). Two preregistered magnitudes were missed (relearning after the second reversal, +0.043 vs. +0.05; long-stay savings +0.167 vs. +0.20) and are recorded as such without changing thresholds. With a true cue, the cue-keyed learner saves most (+0.27) and reaches fitness 0.95, but it depends on the cue being present: with the cue off its fitness is 0.41 in A/B/A and about 0.25 in repeated and long worlds, far below the ordinary learner. We expected a random cue to mislead it; instead it scored 0.96 (hypothesis not supported). A possible reason, *not tested*, is that a changing sentence is itself a change signal that causes a fresh table to be opened. Finally, in this small learnable world the frozen prior added nothing (Laya−none fitness −0.019 for plain, −0.034 for regime), consistent with the earlier finding that the prior has no net value when experience suffices.

5.7 The frozen model did not read environment cues

We tried to let the frozen model supply the missing context signal, i.e. to recognise that two differently worded weather sentences describe the same environment. In a first probe, mean-pooled hidden-state vectors of three states × six training and six evaluation sentences (written before

Prereg	Comparison (held-out dialect, 8 seeds)	Mean [CI]		Verdict
17	A/B/A, no cue: regime−plain savings	+0.215	[+0.164, +0.271]	supported (8/8)
17	A/B/A, true cue: cue−plain savings	+0.345	[+0.304, +0.392]	supported (8/8)
17	A/B/A, true cue: cue−regime savings	+0.126	[+0.089, +0.167]	supported (8/8)
17	A/B/C, no cue: regime fitness non-inferior (> -0.03)	+0.024	[+0.011, +0.038]	supported
17	A/B/A, no cue: regime−plain fitness	+0.084	[+0.062, +0.106]	supported (8/8)
17	A/B/A, random cue: plain fitness − cue fitness ($\geq +0.02$)	−0.159		*not supported* (opposite)
18	half reversal: regime−plain savings	+0.104	[+0.058, +0.153]	supported (8/8)
18	half reversal: regime fitness non-inferior (> -0.03)	+0.005	[−0.016, +0.024]	supported
18	repeated A/B/A/B/A: regime−plain savings	+0.205	[+0.163, +0.253]	supported (8/8)
18	repeated: regime−plain fitness	+0.115	[+0.094, +0.135]	supported (8/8)
18	repeated: regime−plain relearning ($\geq +0.05$)	+0.043		*not supported* (magnitude)
18	long stay A/B/B/B/A: regime−plain savings ($\geq +0.20$)	+0.167	[+0.117, +0.231]	*not supported* (direction yes, magnitude no)

Table 2: Retention results, seed-level bootstrap CI. Sources: `docs/PREREGISTRATION_17.md` §10, `docs/PREREGISTRATION_18.md` §9, `docs/EFFECT_SIZES.md`. In the A/B/A world an ordinary learner's savings are −0.07 (no cue), the environment-table learner's +0.15, the dual learner's +0.03.

looking at results, evaluation wordings with no lexical overlap) gave nearest-centroid accuracy 0.33 (chance 0.33) and cross-wording AUC 0.49; a bag-of-words baseline was no better (0.50, 0.54). In a second probe, five fixed yes/no questions put to the frozen model gave accuracy 0.833 and AUC 0.894, below both pre-set gates (0.9), with a keyword baseline at 0.611 and 0.762 and shuffled features at chance. By the pre-written rules the line was stopped; thresholds were not changed and the sentences were not rewritten to pass. The probes use 18 evaluation sentences and one author-written sentence set, so they are weak evidence; they do not show that no way of reading such cues exists.

5.8 In reproducing populations the individual-level advantage did not carry over

Using the frozen plain and regime genomes unchanged, populations of one innate type reproduce in A/B/A/B/A and A/B′/A worlds (48 worlds per seed, type and pattern, paired world seeds, held-out dialect, 8 seeds; preregistration 19). Establishment means that the population is alive at the horizon with at least 20 births. The environment-table memory that saved old knowledge at the individual level gave no detectable advantage in population establishment, size or recovery. A possible reason, *not tested*, is that persistence is dominated by accumulating enough energy to

Hypothesis	Statistic (threshold)	p	Verdict
H1 repeated: establishment regime−plain	$+0.016$ ($\geq +0.08$)	0.242	not supported
H2 repeated: relative population size	$+2.8\%$ ($\geq +10\%$)	0.188	not supported
H3 repeated: recovery ratio	-0.011 ($\geq +0.05$)	0.770	not supported
H4 half: population size non-inferior (> -1.5)	-0.059	0.0039	supported

Table 3: Population-level results (preregistration 19, `docs/PREREGISTRATION_19.md` §9). Means over 8 seeds: establishment 0.805 (plain) vs. 0.820 (regime) in repeated worlds; population size 9.54 vs. 9.80; per-seed differences have both signs.

reproduce, so differences of a few percentage points in correct feeding are absorbed by food supply and the birth threshold. Descriptive, not confirmatory: in mixed populations the final share of one type was spread from near 0 to near 1 and was no less dispersed than in a neutral control with two labels on one genome (the control used the plain genome rather than the regime genome stated in the protocol; deviation recorded), so drift dominates and mixed shares cannot be read as selection.

5.9 A heritable-mutation probe found no selection direction

In a small development-stage probe (not preregistered; two development genomes per type, 24 worlds per cell, training worlds), a child's genome was the parent's plus Gaussian noise ($\sigma \in \{0, 0.15, 0.5\}$), acquired beliefs not inherited. For each world we took the mean genome of individuals of generation ≥ 2 minus the founder genome, scaled by the noise, and measured the mean pairwise cosine of these displacement vectors across worlds. A value near zero means no common direction. The criterion written beforehand was a value ≥ 0.20 above a permutation null to indicate selection, and < 0.10 to indicate none; all eight cells gave ≤ 0.045 (one slightly above its null). Populations reached about ten generations, so time was not the limit. At $\sigma = 0.5$ mutation itself began to reduce establishment and population size. This indicates no detectable selection direction *in this world and genome representation*; it does not exclude selection elsewhere (`docs/G8B_PROBE.md`).

5.10 Replication on fresh seeds, independent data and more foods

We retrained the plain, environment-table (regime), cue-keyed and a probabilistic "latent-cause" learner on 16 fresh seeds and evaluated once on the UCI Mushroom pool and on the held-out pool, with $K = 6$ and $K = 12$ foods (preregistration 20). A sticky-Chinese-restaurant-process approximation (latent) serves as a comparator for the table-selection rule; it is our simplified approximation, not a full latent-cause model. The retention effect did not shrink on fresh seeds, independent data or twice the foods, and it does not depend on the prior (AUC 0.505). The regime rule also matched or beat the probabilistic latent approximation, whose savings were about zero or negative. The cue-keyed learner scores well with a true cue (fitness .89) and collapses when the cue is off, as before.

5.11 An LLM as the decision-maker did not help; the LLM as a language sensor did

We first let an LLM decide, when a trigger fired, whether to keep, switch or open a belief table, from the individual's own recent meals and each table's expectations (no hidden truth in the prompt). In two development rounds (2916 and 1216 calls; the second used a trigger that fires only when

Hypothesis (16 seeds)	Statistic (threshold)	Verdict
Mushroom, K=6, A/B/A: regime−plain savings	+0.250 ($\geq$ +0.15)	supported, 16/16
Mushroom, A/B/A/B/A: regime−plain savings	+0.205 ($\geq$ +0.15)	supported, 16/16
Mushroom, partial reversal: regime−plain savings	+0.141 ($\geq$ +0.05)	supported, 16/16
Mushroom, K=12, A/B/A: regime−plain savings	+0.365 ($\geq$ +0.10)	supported, 16/16
Mushroom, A/B/A: regime savings not below latent (>-0.05)	+0.131	supported, 16/16
Mushroom, A/B/A: regime−plain fitness ($\geq$ +0.04)	+0.076	supported, 15/16
Held-out pool, A/B/A: regime−plain savings ($\geq$ +0.15)	+0.255	supported, 16/16

Table 4: Preregistration 20 (`docs/PREREGISTRATION_20.md` §11). All seven hypotheses held. Mean fitness / savings on Mushroom, K=6, A/B/A: plain .70/ − .12, regime .77/ + .13, latent .77/ .00, cue-keyed (cue off) .37/ − .52.

error rises) the LLM condition saved $-0.157/-0.171$ and $-0.139/-0.171$ against $-0.050/-0.049$ and $-0.092/+0.030$ for the same trigger with a rule action (two seeds), so by the pre-written rule it gave no sign of benefit; scrambling the meals' outcomes hurt it, so it did read the experience, but it did not read it better than an ordinary learner. Asking instead whether two cue sentences describe the same conditions, the LLM passed all three pre-set gates on 18 held-out wordings (accuracy 1.0, pair AUC 1.0, AUC against unrelated sentences 1.0; Laya's hidden-state vectors scored 0.33/0.49 and its five-question features 0.83/0.89, below the 0.9 gate). A first Bayesian engine with hand-set parameters just missed its preregistered development gate (benefit +0.182 and +0.093 against +0.10), and a review found it was not statistically principled; it was rewritten as described above.

5.12 Bayesian inference with a language sensor, in a world where the wording never repeats

Preregistration 21 trained the benv learner (sensor on) and an ablation with the sensor off on 16 fresh seeds, and compared them with the frozen plain, regime and cue-keyed learners of preregistration 20 in the paraphrase world. With the sensor off the learner is exactly as good as an ordinary learner; the whole gain comes from the language evidence, and shuffling it removes the gain, so the benefit is not "any cue change helps". The cue-keyed learner reaches a similar fitness (every new sentence opens a new table, i.e. it relearns each phase) but not the savings. The weakness is visible: these genomes evolved in a world where the sensor was nearly perfect, and with 30% inverted judgements the advantage is gone.

5.13 Reliability: evolutionary history, lifetime re-estimation and a second channel

Preregistration 22 trained six learners on 8 seeds and evaluated them once over the 16 regime combinations (plus two unseen error rates). A cue-trusting individual evolved in a history where cues were always right collapses when cues mislead (.68 against .91) and, with no cue at all, falls to .26, below the cue-blind .40; one evolved in a mixed history lowers its innate trust, stays at .82

Hypothesis (16 seeds, Mushroom, A/B/A unless noted)	Statistic (threshold)	Verdict
benv−plain savings	+0.353 ($\geq$ +0.15)	supported, 16/16
benv fitness not below regime (>-0.03)	+0.095	supported, 16/16
benv−(sensor off) savings	+0.353 ($\geq$ +0.15)	supported, 16/16
benv−(shuffled sensor) fitness	+0.189 ($\geq$ +0.03)	supported, 16/16
benv with 30% of judgements inverted−plain, fitness	+0.008 ($\geq$ +0.02)	**not supported**
A/B/A/B/A: benv−plain savings	+0.424 ($\geq$ +0.10)	supported, 16/16
Held-out: benv−plain savings	+0.344 ($\geq$ +0.15)	supported, 16/16
benv fitness not below oracle sensor (>-0.03)	−0.0005	supported
benv−cue-keyed savings	+0.176 ($\geq$ +0.08)	supported, 16/16

Table 5: Preregistration 21 (`docs/PREREGISTRATION_21.md` §10). Mean fitness / savings, A/B/A: plain .73/ − .11, regime .81/ + .11, cue-keyed .90/ + .07, benv sensor off .73/ − .11, **benv** .90/ + .25, oracle sensor .90/ + .25, shuffled sensor .71/ − .09, 30% inverted .74/ + .02.

without any cue and generalises to error rates not seen in training, at a cost of one percentage point when the cue is reliable. Re-estimating reliability within a lifetime was, on average, slightly worse than the evolved innate value (7 of 8 seeds negative), contrary to what we predicted. Possible reasons, *not tested*: the numeric evidence already corrects a misleading cue in the posterior, so a fixed moderate trust suffices; our consistency check is noisy and biased early. We cannot exclude other designs in which lifetime updating helps. The second channel helped in 7 of 8 seeds, and the two-channel innate-only learner had the highest mean over the 16 combinations, but the preregistered magnitude was missed by 0.003 and is recorded as not met.

Development-stage notes (not confirmatory). Using LLM-written, often partial-attribute cue sentences, a structured reading (an LLM extracts temperature, wetness, wind, visibility and light, and unknown for what is unsaid; compatibility from likelihood ratios) separated states better than pairwise "same conditions" judgements, but only after we recalibrated a first version that never triggered; sensor-driven switching degraded to the level of an ordinary learner when 30% of phases showed another state's sentence; and a 40-tick thinking delay (the individual rests while the sensor "thinks") kept most of the gain in a world with 1200-tick phases.

6 Discussion

Where the frozen prior pays. Across protocols that differ in design, the prior's fitness value ordered itself as learnable world (−0.005), immediate feedback (+0.015), delay 40 (+0.049) and delay 80 (+0.073); this ordering is descriptive across different preregistered protocols, not one confirmatory test. A delay-aware credit assignment shrinks the delay advantage to +0.017 (interval including zero), so the finding is the prior's value relative to a weak learner under delay.

Retention is a property of the memory structure. An environment-table memory turns the ordinary learner's negative savings positive, and this held on fresh seeds, on an independent data source where the prior is uninformative, and with more foods. The credit belongs to the innate memory structure, not to the frozen model.

Hypothesis (8 seeds, Mushroom)	Statistic (threshold)	Verdict
Misleading cue, one channel: mixed-history−clean-history fitness	+0.243 ($\geq$ +0.04)	supported, 8/8
Reliable cue: cost of suspicion (>-0.03)	−0.009	supported
Reliable cue: cue user−cue-blind fitness	+0.572 ($\geq$ +0.20)	supported, 8/8
All 16 combinations: worst seed-wise gap to cue-blind (>-0.05)	+0.255 (mean of minima)	supported
Lifetime re-estimation−innate trust only	−0.031 ($\geq$ +0.02)	**not supported (opposite)**
Reliable cue: re-estimation harmless (>-0.03)	−0.009	supported
Second channel: two-channel−one-channel (re-estimating)	+0.047 ($\geq$ +0.05)	**significant but below threshold**
Unseen error rates 0.15 / 0.45: cue user−cue-blind	+0.528 ($\geq$ +0.05)	supported, 8/8
Held-out pool, reliable cue	+0.572 ($\geq$ +0.20)	supported, 8/8

Table 6: Preregistration 22 (`docs/PREREGISTRATION_22.md` §10). Mean fitness over the 16 combinations: cue-blind .39, one channel innate-only .91, one channel evolved in a clean history .67, one channel with re-estimation .88, two channels innate-only .93, two channels with re-estimation .90. Evolved innate trust in a cue: 0.99 (clean history) against 0.91 (mixed history).

Language as evidence, not as a command. Letting an LLM decide when to switch memory gave no benefit, and it was not better than a rule; letting it judge whether two cue sentences describe the same conditions and fusing that as a noisy sensor into a proper posterior gave the largest retention we measured, close to an oracle sensor. The distinction matters: the same model is a poor sequential reasoner about noisy meals but a good judge of paraphrase, and the architecture should use it for the second.

Who is responsible for cue reliability. We expected a division of labour in which evolution sets the innate trust, lifetime learning calibrates it where it varies across worlds, and redundancy cross-checks channels. Evolution did its part decisively: the innate trust must be evolved in a history that contains unreliable cues. Lifetime re-estimation did not help in our design, and redundancy helped but fell just short of the preregistered size. We therefore do not claim a three-way division; we claim that the evolutionary history decides the innate trust, and that the numeric cross-check inside Bayesian inference already protects against a misleading cue.

What did not follow. The individual-level advantage did not become a detectable population-level advantage, and a mutation probe found no selection direction; whether this reflects the simplified reproduction, the degraded calibration, small populations or a genuine lack of selection pressure we cannot tell. We also have no evidence that the frozen Laya model understands environmental change: two probes of its hidden states and questions did not pass.

7 Limitations

- Dialects are written by one author; the controller thresholds were chosen in development; the held-out phrase pool was evaluated repeatedly (prereg 02, 03, 04; only aggregates inspected, not used for tuning); the fact-dialect pool was evaluated once.

- Small seed counts (6–8); intervals are across-seed percentile bootstrap.
- Templated food scenes; the prior is used only for "edible or not". Simulation only; not a claim about real organisms or open worlds.
- Laya−none contrasts contain a "can read vs. cannot read" component (no-prior agents ignore external cues), so they are not fair learning comparisons.
- Laya is a 421M-parameter model with cached priors; throughput is not extrapolated.
- The cross-protocol ladder in the discussion is post hoc.
- Reversal worlds have six foods and one or two reversals; the environment-table learner holds at most four tables, so table-full behaviour is an implementation detail, not a memory capacity of living organisms. Genomes were trained in three-phase worlds, so longer or more frequent reversals are out-of-distribution tests.
- The reproducing-population parameters come from a degraded calibration; reproduction is a fixed instinct; the long-stay population world was not evaluable (establishment 0 in development) and mixed-population shares are dominated by drift; the mutation probe has two development seeds and one world.
- Several explanations (random-cue success, collapse without cues, absence of a population effect, why lifetime re-estimation did not help) are hypotheses that were not tested.
- The language sensor is an LLM's cached judgement of 36 (and, for the second channel, 36 more) cue sentences; the sentences are written by the authors or by the same LLM, the two states of the dense world are simple, and the second channel's independence is independence of the injected faults, not of the information source. We did not test other LLMs or prompts. The cue-keyed learner was trained with a true cue and evaluated with paraphrased cues, so that comparison is out of distribution for it.
- The probabilistic latent-cause comparator is a simplified online approximation; we did not compare with neural continual-learning methods (such as elastic weight consolidation). In this world beliefs are per-description tables without shared weights, so protecting important weights corresponds to lowering forgetting for well-evidenced entries, which the plain learner's forgetting gene already covers; that argument is not an experiment.
- Eight seeds in preregistration 22 make Holm's correction severe; the Mushroom descriptions are a template over real attributes, not natural text; the held-out pool has been evaluated many times.
- The held-out dialect pool was evaluated by preregistrations 17–19 as well; none of these results were used to tune training.

8 Reproducibility and Disclosure

Code is released under Apache-2.0 and data (dialects, word pools, experiment outputs, documents) under CC BY 4.0; the Laya weights are not redistributed (Apache-2.0 upstream, revision recorded in the repository). Each confirmatory result has a frozen protocol (`docs/PREREGISTRATION_02` to `22`) whose git commit precedes training and evaluation, genome hashes committed before evaluation, and raw outputs under `experiments/`; `REPRODUCE.md` lists the commands. The cached Laya

priors make evaluation reproducible without the model. Deviations are recorded in each protocol's deviations section, including one interrupted-and-rerun evaluation (preregistration 18), one control that used a different genome than the protocol stated (preregistration 19), and a naming clash in preregistration 21: while writing the new learner we accidentally overwrote the source file of an earlier learner (preregistration 14); we restored it from version control byte for byte before any evaluation used it and renamed the new learner, changing no hypothesis or threshold. Costs of the LLM calls (about 10^4 distinct calls in total for the probes and the sensor matrices) were not limited by a budget.

AI assistance. The simulator code, experiment scripts, analysis and a draft of this text were produced with substantial assistance from an AI coding assistant (Claude, Anthropic) working under the author's direction; the author set the research questions and decisions, reviewed protocols and results, and is responsible for the content. All numbers trace to the repository's raw results. References were checked at the metadata level against publisher or arXiv pages; the specific claims attributed to each were checked against abstracts or search summaries, not full texts (`refs.bib` records the status per entry).

Funding and interests. No funding; no competing interests.

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